Edge Detection using Sobel Filter

$$\begin{bmatrix} 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \end{bmatrix}_{5\times5}$$

$$\text{Outputsize} = \frac{N-F+2P}{S}+1$$

$$= \frac{5-3+0}{1}+1$$

$$= \underline{\underline{3}}$$

where
$N \Rightarrow$ Input image size $=5$
$F \Rightarrow$ Filter size $=3$
$P \Rightarrow$ Padding $=0$
$S =$ Stride $=1$

$\therefore$

Output size $= \underline{\underline{3\times3}}$

$$\text{Sobel } X = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

1st Convolution:

$$\text{Image Patch} = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix} ; \text{ Sobel } X = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\text{Output}(1,1) = (-10+0+10) + (-20+0+20) + (-10+0+10)$$

$$= \underline{\underline{0}}$$

Output (1,2) :

$$\begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\text{Output}(1,2) = (-1\times10) + (0) + (80) + (-20 + 0 + 160) + (-10 + 0 + 80) = 280$$

Output (1,2) = 280.

Output (1,3);

$$\begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\text{Output}(1,3) = (-10 + 0 + 80) + (-20 + 0 + 160) + (-10 + 0 + 80)$$

Output (1,3) = 280

As all the rows are identical;

The remaining rows will give same answer

Final Output = $\begin{bmatrix} 0 & 280 & 280 \\ 0 & 280 & 280 \\ 0 & 280 & 280 \end{bmatrix}$

Analysis:

The Sobel-X operator detects vertical edges, The given image intensity changes from 10 to 80 btw 3rd and 4th coloumns, producing a strong response of 280. Homogenous regions produce value close to zero because positive and negative weights cancel each other.